LLM-Powered Academic Research Planning Agent

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1. Introduction and Scope

Systematic reviews require researchers to screen large volumes of candidate abstracts before synthesis begins. This report specifies an LLM-powered planning agent for a health-sciences research group that automates search, retrieval and first-pass screening, leaving inclusion decisions to a researcher.

2. System Requirements and Assumptions

2.1 Functional requirements

- The system will accept a health-sciences research question and automatically generate appropriate literature-search queries.
- It will retrieve academic records from PubMed/PubMed Central and Semantic Scholar, including titles, abstracts, authors, publication dates and identifiers such as DOI or PMID.
- Retrieved records will be deduplicated, filtered and ranked according to relevance and predefined inclusion/exclusion criteria.
- Relevant studies will be summarised into a structured evidence record containing the source, supporting extract, screening decision and publication status. Search queries, filters and timestamps will also be logged to support reproducibility.
- A human researcher will verify evidence before final acceptance. The prototype supports academic research only. It will not process patient-identifiable data, provide clinical recommendations or replace researcher judgement.
- The prototype will primarily process English-language literature. Full-text retrieval will be limited to legitimately available open-access material, for example through PubMed Central.

2.2 Success criteria mapping

The proposed system directly addresses the assignment criteria. Retrieval is achieved through academic APIs. Processing is performed through deduplication, screening, ranking and LLM-based summarisation. Storage and presentation are provided

through a structured SQLite evidence store and Streamlit interface. Traceable source information is retained for each accepted result.

- Retrieval quality: recall of at least 0.80 against a hand-screened sample of 50 abstracts.
- Traceability: every claim resolves to an identifier and supporting extract, or validation rejects it.
- Fidelity: a 20-record audit finds no claim absent from its abstract.
- Storage: store contents reconcile with the reviewer decision log.

2.3 Key assumptions

- PubMed provides metadata and abstracts; full text comes through PubMed Central only, and not for every record.
- Gemini 3.6 Flash on the Gemini API free tier supplies LLM capability throughout; no paid endpoint is assumed anywhere in the design.
- Three genuine researcher needs are out of scope: wider publisher coverage, citation-graph exploration, continuous monitoring.

3. Design Decisions and Architecture

- **Hybrid architecture, not purely reactive or purely deliberative.** A reactive design cannot maintain ordered research subgoals, and classical symbolic planning needs predefined representations ill-suited to unstructured literature. Layering the two puts deliberation at decomposition and reaction at retrieval (Wooldridge, 2009).

- **BDI-inspired planning state, a conceptual model, not a formal interpreter.** Beliefs are the records retrieved and the current search state; desires are the research objective; intentions are the committed subtask queue (Rao and Georgeff, 1995). Committing to that queue bounds further deliberation, the role Bratman, Israel and Pollack (1988) assign to plans in a resource-bounded agent.
- **ReAct as the execution loop**, alternating reasoning with action rather than planning the whole search in advance (Yao et al., 2023).
- **Two cooperating agents, not one per operation.** The Planning and Retrieval Agent decides which subgoal to pursue, when one has yielded enough to drop, and when to re-plan. The Curation Agent filters, ranks and summarises; deduplication, validation and storage stay outside the LLM, limiting variability. This draws on LEADS's decomposition of literature mining into search, eligibility assessment and extraction (Wang, Z. et al., 2025), here consolidated to reduce orchestration overhead.
- **Abstracts and open-access full text only**, since most journal content remains paywalled.
- **A schema-validated JSON buffer, not an agent communication language.** KQML suits autonomous, heterogeneous agents negotiating across independently controlled knowledge bases (Finin et al., 1994). Here both agents are one trusted application with predefined roles and a shared schema, so a validated buffer coordinates them without that overhead.
- **A human verification gate before storage.** Wang, G. et al. (2026) recommend traceable evidence chains and verification steps to reduce the expert's vigilance burden, so every record carries its extract.

4. Graphical System Designs

Figure 1 shows the structure, Figure 2 the behaviour.

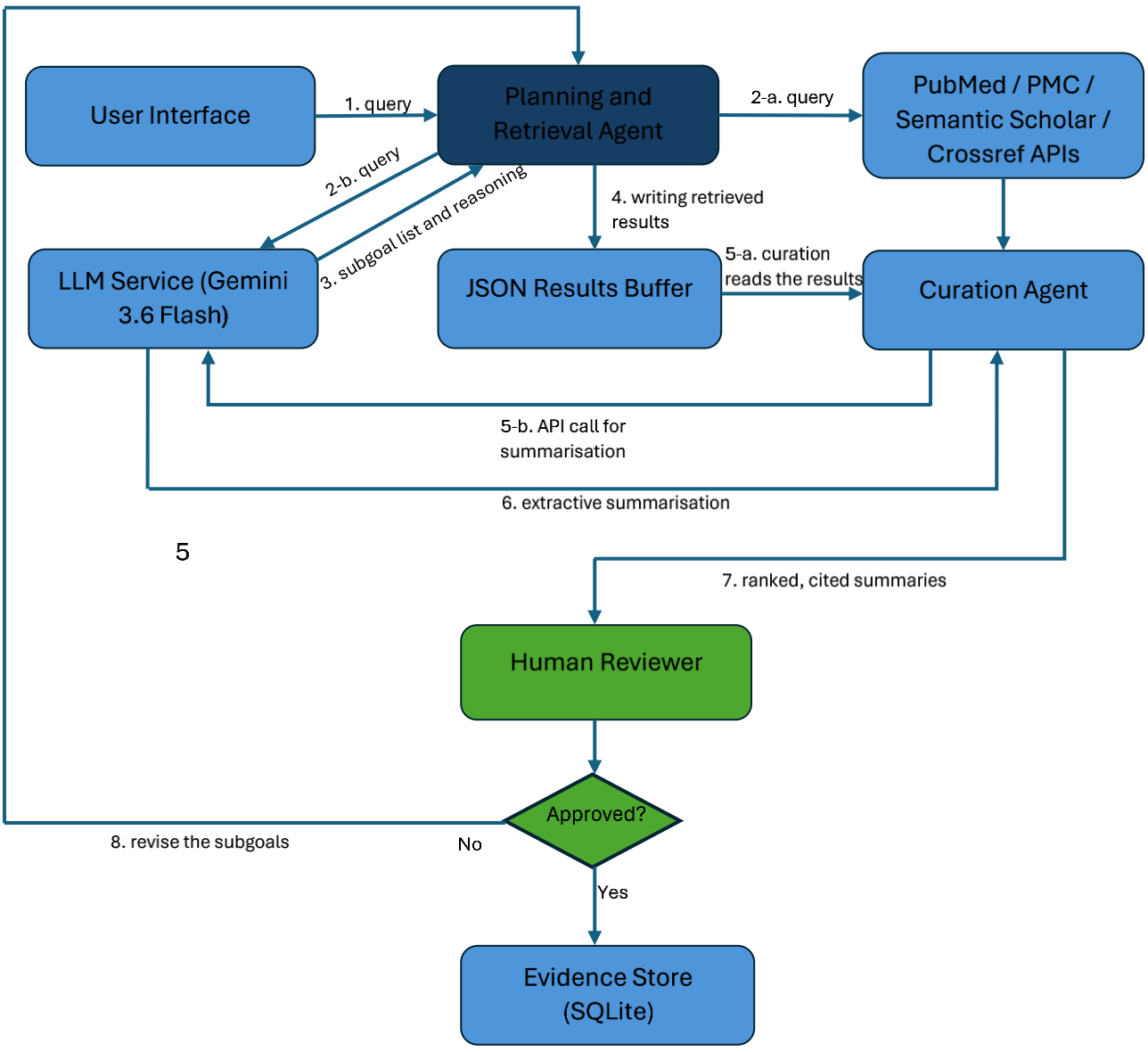


Figure 1: Two-agent system overview.

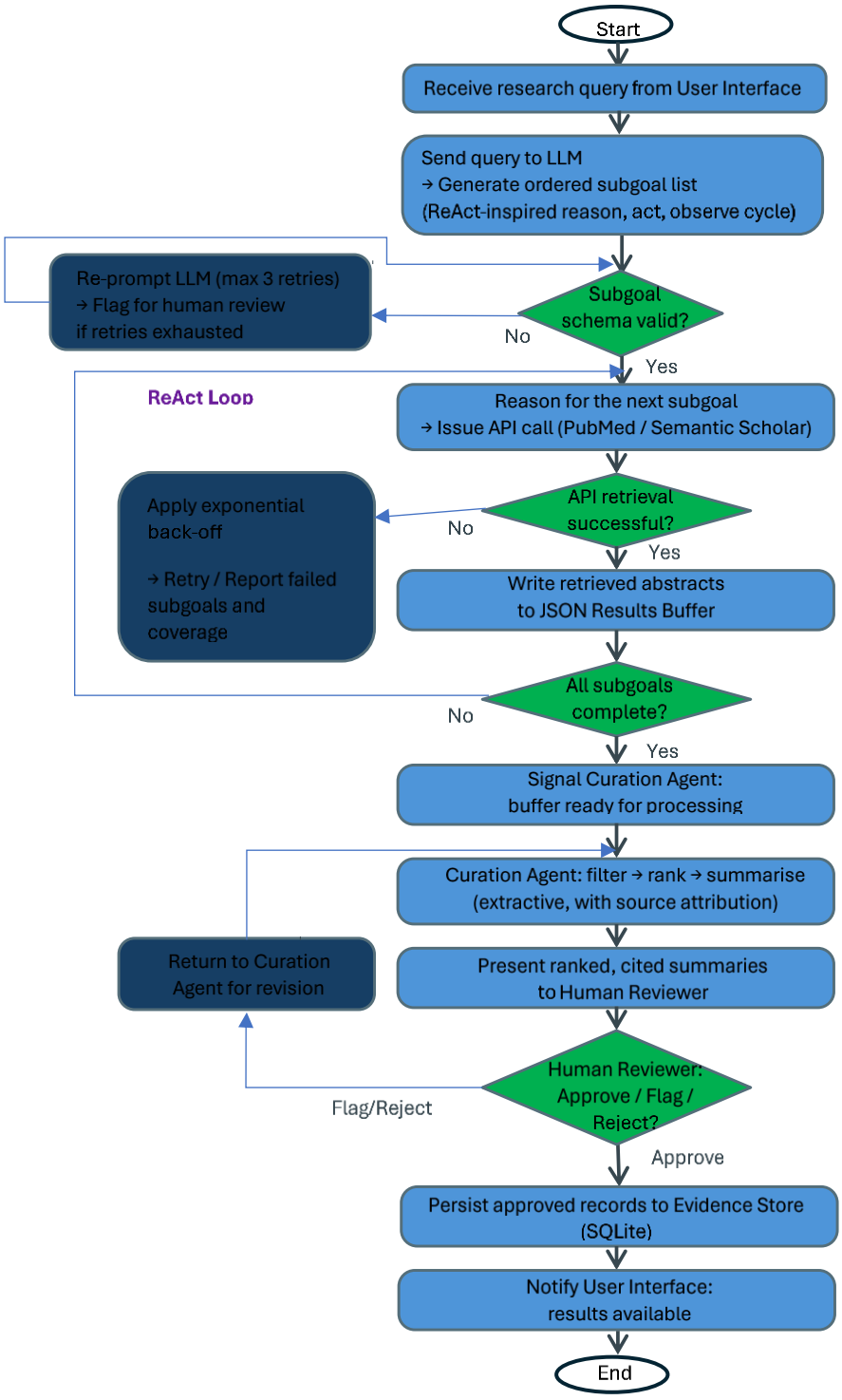


Figure 2: Request lifecycle activity view.

5. Development Approach

Development is incremental across three two-week sprints: the Planning and Retrieval Agent, then the Curation Agent and interface, then integration and testing. This is preferred to big-design-up-front because agent development remains experimental, with no settled methodology (Wooldridge, 2009); each sprint closes with a runnable slice.

6. Anticipated Challenges

- **Decomposition reliability.** Subgoal lists are generated, not guaranteed, so each is schema-validated with three re-prompts before escalation. Bounded retries matter because a resource-bounded agent must stop deliberating and act (Bratman, Israel and Pollack, 1988).
- **Partial observability.** Abstract-only retrieval leaves the Curation Agent reasoning over an incomplete view of each study, an inaccessible environment in Wooldridge's (2009) terms, so every record is labelled abstract-derived.
- **Hallucination and misattribution.** Summarisation is extractive, and every claim carries its supporting extract and identifier, making fabrication visible rather than merely unlikely (Wang, Z. et al., 2025; Wang, G. et al., 2026).
- **Silent degradation.** The harder failure is not a crash but a partial retrieval that still returns a polished summary set, hiding what is missing. Each run reports failed subgoals and coverage, the account PRISMA requires of any automation used in screening (Page et al., 2021).
- **Limits of a single gate.** Verification is not free: re-reading every extract moves reviewer effort rather than removing it. The published gain is modest, 0.81 recall

against 0.78 unaided (Wang, Z. et al., 2025), so the aim is a better hour of screening, not automation.

- **Legal and ethical exposure.** Retrieval is confined to metadata, abstracts and licensed open-access text, so publisher licences are respected rather than tested, and Crossref supplies retraction status. Published health literature is de-identified at source but can carry patient-level detail. Retrieved text is not re-screened, so data flows are documented against GDPR and review-board questions.

7. Tools, Libraries, Models and Environments

Figure 3 records the alternatives considered.

- Orchestration: Python, LangChain and LangGraph for agent workflows and tool use.
- LLM: Gemini 3.6 Flash on the Gemini API free tier, kept replaceable behind the orchestration layer (Google DeepMind, 2026).
- Output validation: Pydantic, validating structured model outputs.
- Literature and publication-status metadata: PubMed/PMC E-utilities, Semantic Scholar, Crossref.
- Processing and deduplication: pandas, RapidFuzz.
- Persistence and interface: SQLite for free local storage, Streamlit for the user interface.
- Development and testing: VS Code, GitHub, pytest.

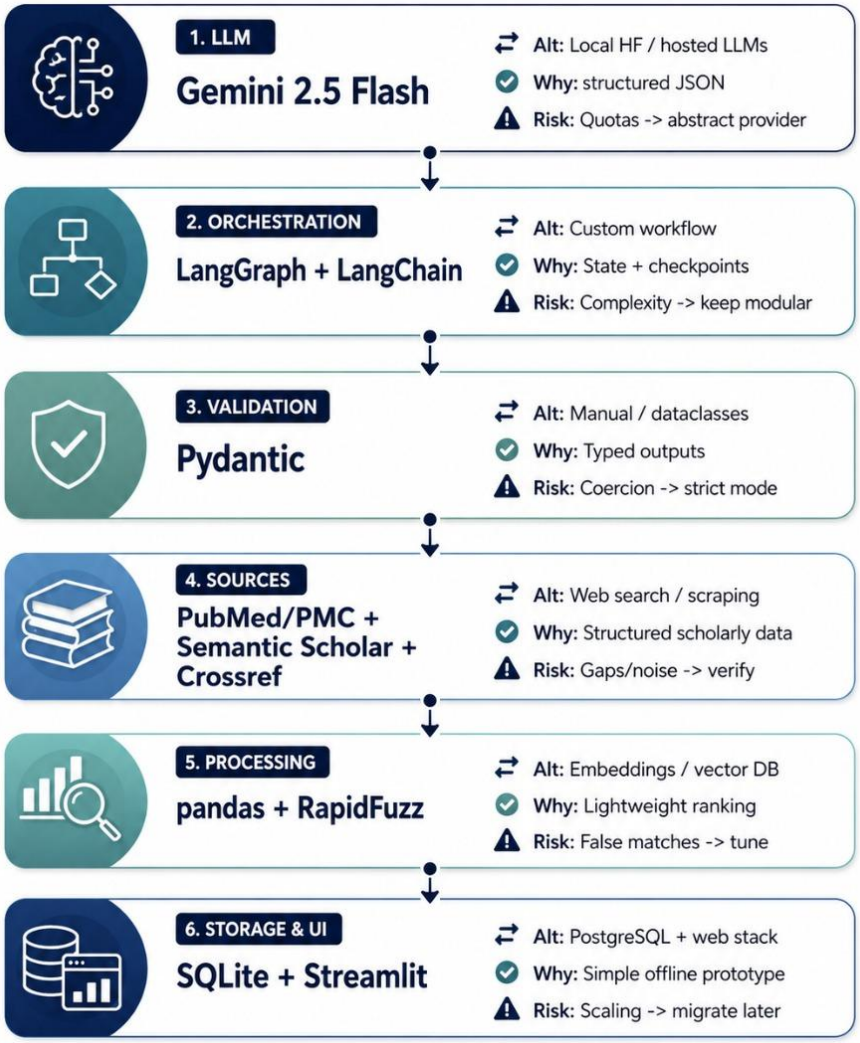


Figure 3: Technology decisions, alternatives and tooling risks.

8. Conclusion

This is a two-agent hybrid system, justified against reactive and symbolic alternatives and against a full agent communication language, whose review gate targets the failure that matters most: a fabricated citation entering the evidence base. The residual risk is coverage, since a two-source agent will miss evidence.

References

Bratman, M.E., Israel, D.J. and Pollack, M.E. (1988) 'Plans and resource-bounded practical reasoning', *Computational Intelligence*, 4(3), pp. 349-355.

Finin, T. *et al.* (1994) 'KQML as an Agent Communication Language', *Proceedings of the Third International Conference on Information and Knowledge Management (CIKM '94)*, Gaithersburg, MD. New York: ACM, pp. 456-463. Available at: https://doi.org/10.1145/191246.191322

Google DeepMind (2026) Gemini 3.6 Flash: model card. Available at: https://deepmind.google/models/model-cards/gemini-3-6-flash/ (Accessed: 27 August 2026).

Page, M.J. *et al.* (2021) 'The PRISMA 2020 statement: An updated guideline for reporting systematic reviews', *International Journal of Surgery*, 88, 105906. Available at: https://doi.org/10.1016/j.ijsu.2021.105906

Rao, A.S. and Georgeff, M.P. (1995) 'BDI Agents: From Theory to Practice', *Proceedings of the First International Conference on Multiagent Systems (ICMAS-95)*, San Francisco, CA, 12-14 June. Menlo Park, CA: AAAI Press, pp. 312-319.

Wang, G. *et al.* (2026) 'Human–large language model collaboration in clinical medicine: a systematic review and meta-analysis', *npj Digital Medicine*, 9(1), 195. Available at: https://doi.org/10.1038/s41746-026-02382-2

Wang, Z. *et al.* (2025) 'A foundation model for human-AI collaboration in medical literature mining', *Nature Communications*, 16(1), 8361. Available at: https://doi.org/10.1038/s41467-025-62058-5

Wooldridge, M. (2009) *An Introduction to MultiAgent Systems*. 2nd edn. Chichester: John Wiley & Sons.

Yao, S. *et al.* (2023) 'ReAct: Synergizing Reasoning and Acting in Language Models', *11th International Conference on Learning Representations (ICLR 2023)*, Kigali, Rwanda, 1-5 May. Available at: https://doi.org/10.48550/arXiv.2210.03629

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